

**NAME: MUHAMMAD MOHSIN**

**ROLL NO: 22F-BSAI-110**

## **DEEP LEARNING ASSIGNMENT REPORT**

### **Can AI Save Lives? Building an Intelligent Clinical Early Warning System**

#### **Section 1: The Clinical Problem**

Patient deterioration prediction is difficult because clinical decline is rarely sudden or obvious. In many hospital cases, a patient's condition worsens gradually through small changes in heart rate, blood pressure, oxygen saturation, respiratory rate, temperature, laboratory values, and clinician observations. Unlike a standard classification task where the input may be fixed and clearly separated, deterioration prediction involves noisy, incomplete, imbalanced, and time-dependent data. In the PhysioNet Sepsis Prediction Challenge dataset used in this project, the positive deterioration/sepsis class was much smaller than the stable class, which made the task clinically and technically challenging.

Tabular vital-sign data is useful because it captures measurable physiological signals, but it is not sufficient on its own. Numbers may show that a patient has tachycardia, fever, or hypotension, but they do not fully explain the clinical context. Free-text clinical notes can add information such as "patient appears confused," "worsening respiratory distress," "poor urine output," or "concern for infection." These phrases may signal deterioration before a single numeric threshold becomes extreme. Because the selected PhysioNet dataset does not contain real clinical notes, synthetic clinical notes were generated from vital-sign abnormalities to simulate the Transformer-based language model section.

Recall matters more than accuracy in this application because missing a true deterioration case is far more dangerous than raising a false alarm. For example, if a patient develops fever, low blood pressure, high respiratory rate, and low oxygen saturation, but the model predicts that the patient is stable, the clinical team may not intervene quickly. This false negative could delay antibiotics, fluids, ICU transfer, or closer monitoring. A false positive may create extra workload, but a false negative may cost a patient's life. Therefore, the model evaluation focused strongly on recall for the risk class rather than accuracy alone.

## Section 2: Baseline Design Decisions

The first generation model used a shallow deep neural network as a baseline. This model treated each hourly clinical record as an independent snapshot. Missing values were handled using median imputation, features were normalized using standard scaling, and columns with more than 95 percent missing values were removed. Class weighting was used because only a small portion of the dataset represented sepsis or deterioration cases.

One important issue in deep learning is the vanishing gradient problem. In deep networks, gradients can become very small as they are propagated backward through many layers. When this happens, early layers learn very slowly, and the network may fail to capture useful patterns. ReLU activation was used because it reduces this problem compared to sigmoid or tanh activations. Batch Normalization was also added because it stabilizes the distribution of layer inputs, supports faster convergence, and helps the model train more reliably.

The DNN was trained using two optimizers: Adam and SGD. Adam achieved better overall accuracy and F1-score, while SGD trained faster and produced slightly higher recall. Adam adapts the learning rate for each parameter and usually converges more smoothly, while SGD with momentum is simpler and computationally lighter but may require more careful tuning. In this experiment, Adam was selected as the final DNN baseline because it gave a stronger overall balance of metrics.

A loss function measures the error for one prediction or one batch during training, while a cost function usually refers to the overall average objective minimized across the dataset. Binary cross-entropy was used because this is a binary classification problem: stable patient versus deterioration risk. Dropout was added as a regularization technique. It randomly disables some neurons during training, which prevents the network from relying too heavily on specific features and reduces overfitting.

## Section 3: Modelling Patient Timelines

The second generation models used recurrent architectures to capture patient timelines. Instead of treating each row as a separate snapshot, the data was restructured into 12-hour patient windows. This is more clinically realistic because deterioration is often a story that unfolds across time. A patient's heart rate, oxygen saturation, temperature, blood pressure, and respiratory rate may slowly worsen before a severe event occurs.

RNNs can suffer from the vanishing gradient problem even more severely than standard feedforward networks because information must be passed across many time steps. If gradients shrink over time, the model may forget earlier clinical signals that are still important. LSTM networks address this issue using gates: the input gate controls what new information is stored, the forget gate controls what information is removed, and the output gate controls what information is passed forward. This allows LSTMs to preserve important signals over longer sequences.

GRU models simplify the LSTM design by combining some gate mechanisms. This makes GRUs faster and lighter, which can be useful in real-time clinical monitoring where latency and computational cost matter. However, in this experiment, GRU produced many false positives and a lower F1-score than LSTM. The LSTM achieved the best balance, with accuracy of 0.9000, recall of 0.5714, and F1-score of 0.5000.

The Bi-LSTM did not improve performance in this experiment. More importantly, bidirectional LSTM is not appropriate for a live ICU early warning system because it uses both past and future context. In real-time monitoring, future measurements are not available at the moment of prediction. Bi-LSTM may be useful for retrospective analysis, research, or offline chart review, but a unidirectional LSTM is more appropriate for deployment because it only uses information available up to the current time.

## Section 4: The Transformer and Clinical Language

Pre-trained language models are useful because they bring knowledge learned from large text corpora into a smaller clinical task. In the context of ClinicalBERT, transfer learning means that the model has already learned medical vocabulary, clinical phrasing, abbreviation patterns, and relationships between healthcare terms from large biomedical or clinical text sources. Training a language model from scratch on a small assignment dataset would not provide enough data for meaningful language understanding.

Self-attention is valuable because clinical notes are not always read in a simple left-to-right manner. A doctor may scan for key warning terms such as fever, hypotension, tachypnea, altered mental status, infection, or oxygen desaturation. Transformer models use self-attention to connect important words across the note and identify which terms contribute most to the classification decision.

Since the PhysioNet Sepsis dataset does not include real free-text notes, synthetic notes were generated from vital-sign abnormalities. The simulated attention-style visualization highlighted terms such as fever, hypotension, tachypnea, and tachycardia. These are clinically meaningful warning signs and align with what a clinician would expect in sepsis or deterioration risk. This suggests that the text model focused on medically relevant terms rather than unrelated words.

Two fine-tuning strategies were compared. The frozen simulation represented using a pre-trained model as a fixed feature extractor with only the classification head trained. This approach is faster, cheaper, and safer when the dataset is small. The full fine-tuning simulation represented updating the full model. It achieved slightly better accuracy, precision, and F1-score, but in a real Transformer this would require more GPU memory, more training time, and greater risk of overfitting. Frozen fine-tuning is preferable when the dataset is small or computational resources are limited. Full fine-tuning is justified when there is enough labeled clinical text and the deployment benefit outweighs the training cost.

Positional encoding matters because Transformers do not naturally process word order like RNNs. Clinical meaning can depend on sequence and context. For example, “no fever” and “fever worsening overnight” have very different meanings. Positional information helps the

model understand where terms appear in the note and how they relate to surrounding words. Transformers also support parallel processing, which is important for hospital-scale systems that may need to process thousands of notes per hour.

## Section 5: Deployment Verdict

Based on the experimental results, the LSTM model is the recommended deployment candidate for a real ICU early warning system. It achieved the best overall balance between accuracy, recall, F1-score, and real-time usability. The DNN baseline was simpler but ignored patient timeline patterns. The GRU was faster but produced too many false positives. The Bi-LSTM was not suitable for live monitoring because it depends on future context. The simulated ClinicalBERT models were useful for explanation and note analysis, but the text component was synthetic because the selected dataset did not include real clinical notes.

In a real deployment, the safest architecture would combine a unidirectional LSTM for real-time vital-sign monitoring with a ClinicalBERT-style note model once real clinical notes are available. This would allow the system to use both structured physiological trends and unstructured clinician observations. The final output should not replace doctors. It should act as a decision-support system that highlights high-risk patients and explains the warning signals.

Several ethical concerns must be addressed before clinical deployment. Bias is a major risk because models trained on one hospital population may not perform equally well across age groups, genders, ethnic groups, or care settings. Privacy is also critical because medical data is highly sensitive and must be protected through secure storage, access control, and de-identification where appropriate. Accountability must also be clear. If the model misses a deteriorating patient or raises repeated false alarms, the hospital must define who is responsible for reviewing and acting on the alert.

If the hospital later wanted to analyze chest X-rays alongside clinical notes and vitals, the architecture would need to become multimodal. A CNN or Vision Transformer could process chest X-ray images, ClinicalBERT could process notes, and LSTM could process time-series vitals. The outputs from all three branches could be fused into a final classification layer that predicts deterioration risk. This would create a richer early warning system, but it would also require stronger validation, explainability, and clinical governance before deployment.